United States Geological Survey

Reference Material Information Sheet

BHVO-2 (Hawaiian Volcano Observatory Basalt)

Introduction

Material used in the preparation of BHVO-2 was collected from the surface layer of the pahoehoe lava that overflowed from the Halemaumau Crater in Hawaii's Kilauea Caldera in the fall of 1919. This is the same location used to provide material for BHVO-1 (Flanagan, 1976). Collection of BHVO-2 was accomplished in 1995 under the direction of Carl Thornber, USGS.

The USGS has not published a metrologically traceable characterization of the elemental content of this material. The values presented in Table 1 and Table 2 are from a published study that summarizes multi-laboratory analyses of mass fraction values for elements in the material. It is recommended that the user of this material consult the reference provided to determine the estimated value(s) that are the best fit for their purpose.

Expiry

The material is considered geochemically stable with proper storage.

Contact

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Table 1

Estimates of major oxide concentrations in BHVO-2 from published literature

Oxide	Mass Fraction (g/100 g)	Uncertainty (g/100 g)	n[1]	Notes
Al_2O_3	13.44	0.06	*24*	*2,4*
CaO	11.40	0.06	*28*	*2,4*
Fe_2O_3T	12.39	0.09	*25*	*2,4,5*
K_2O	0.5130	0.0037	*25*	*2,4*
MgO	7.257	0.042	*26*	*2,4*
MnO	0.1690	0.0019	*35*	*2,4*
Na_2O	2.219	0.048	*27*	*2,4*
P_2O_5	0.2685	0.0050	*27*	*2,4*
SiO_2	49.60	0.14	*27*	*2,4*
TiO_2	2.731	0.018	*50*	*2,4*

Table 2

Estimates of trace element concentrations in BHVO-2 from published literature

Element	Mass Fraction (mg/kg)	Uncertainty (mg/kg)	n[1]	Notes
Ba	130.9	1	*125*	*3,4*
Be	1.076	0.046	*33*	*3,4,6*
Bi	0.0148	0.0043	*12*	*3,4,6*
Cd	0.152	0.049	*23*	*3,4*
Ce	37.53	0.19	*134*	*3,4*
Co	44.89	0.32	*102*	*3,4*
Cr	287.2	3.1	*93*	*3,4*
Cs	0.0996	0.0022	*92*	*3,4,6*
Cu	129.3	1.4	*92*	*3,4*
Dy	5.280	0.028	*130*	*3,4*
Er	2.511	0.014	*128*	*3,4*
Eu	2.043	0.012	*132*	*3,4*
Ga	21.37	0.20	*74*	*3,4*
Gd	6.207	0.038	*130*	*3,4*
Hf	4.470	0.025	*6*	*3,4,7*
Ho	0.9887	0.0053	*130*	*3,4*
In	0.117	0.045	*7*	*3,4*
La	15.20	0.08	*132*	*3,4*
Li	4.500	0.085	*66*	*3,4*
Lu	0.2754	0.0024	*7*	*3,4,7*
Mo	4.07	0.16	*31*	*3,4*
Nb	18.10	0.20	*121*	*3,4*
Nd	24.27	0.25	*8*	*3,4,7*
Ni	119.8	1.2	*105*	*3,4*
Pb	1.653	0.038	*100*	*3,4*
Pr	5.339	0.028	*126*	*3,4*
Rb	9.261	0.096	*119*	*3,4*
Sb	0.1034	0.0079	*26*	*3,4,6*
Sc	31.83	0.34	*96*	*3,4*
Sm	6.023	0.057	*9*	*3,4,7*
Sn	1.776	0.059	*34*	*3,4,6*
Sr	394.1	1.7	*123*	*3,4*
Ta	1.154	0.019	*114*	*3,4*
Tb	0.9392	0.0060	*130*	*3,4*
Th	1.224	0.016	*124*	*3,4*
Tl	0.0224	0.0015	*28*	*3,4*
Tm	0.3349	0.0031	*104*	*3,4*
U	0.412	0.035	*23*	*3,4*
V	318.2	2.3	*101*	*3,4*
W	0.251	0.035	*23*	*3,4*
Y	25.91	0.28	*127*	*3,4*
Yb	1.994	0.027	*132*	*3,4*
Zn	103.9	1.0	*90*	*3,4*
Zr	171.2	1.3	*121*	*3,4*

Notes for Table 1 and Table 2

The number of decimal places in mass fraction and uncertainty values are as given in the citation.

(1) n is the number of laboratory mean estimates that were used to calculate the overall mean and uncertainty of the estimate reported in the cited publication. See the cited publication for details.
(2) Estimate from Table 3 of Jochum et al., 2016.
(3) Estimate from Table 4 of Jochum et al., 2016.
(4) Uncertainty is expressed as 95% confidence level.
(5) Fe_2O_3T is total iron expressed as Fe_2O_3.
(6) As reported by Jochum et al. (2016), measurements were all made by a single technique (inductively coupled plasma-mass spectrometry).
(7) As reported by Jochum et al. (2016), measurements were made by isotope dilution-mass spectrometry.

References

Flanagan, F.J., 1976, Descriptions and analysis of eight new USGS rock standards: U.S. Geological Survey Professional Paper 840, 192 p. [Also available at https://doi.org/10.3133/pp840.]

Jochum, K.P., Weis, U., Schwager, B., Stoll, B., Wilson, S.A., Haug, G.H., Meinrat, A.O., and Enzweiler, J., 2016, Reference values following ISO guidelines for frequently requested rock reference materials: Geostandards and Geoanalytical Research, v. 40, p. 333-350. [Also available at https://doi.org/10.1111/j.1751-908X.2015.00392.x.]